Introduction

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Introduction

During natural disasters such as typhoons and severe floods, real-time social media streams carry millions of messages about rescue needs, infrastructure damage, and relief offers (Imran et al., 2016). This volume outpaces manual review, so extracting critical information depends on automated natural language processing (NLP). This study formulates disaster triage as three concurrent NLP tasks. These comprise token-level named entity recognition (NER) for locations and infrastructure, sequence-level intent classification for humanitarian needs (Alam et al., 2021), and sequence-level urgency prioritization of crisis messages (Wang et al., 2021).

A common approach trains a separate single-task learning (STL) model for each prediction target (Ruder, 2017). Latency and memory consumption then grow linearly with task count ($\mathcal{O}(K)$ for K tasks) because each model processes every input in a separate forward pass. In contrast, hard-parameter-sharing multi-task learning (MTL) uses a single multilingual Transformer backbone to generate shared representations in a single forward pass, achieving near-constant inference latency ($\mathcal{O}(1)$ relative to K). However, joint multi-task training couples heterogeneous loss functions, producing two primary failure modes during backpropagation, namely gradient magnitude disparities and directional conflicts (Z. Chen et al., 2018; Yu et al., 2020). Yet on low-resource, code-switched text, joint training helped only under specific task pairings, training orders, and data sizes (Adouane & Bernardy, 2020).

To resolve these computational and optimization challenges, this study develops an empirical benchmarking framework grounded in the Design Science Research

Methodology (DSRM). The investigation benchmarks candidate multi-task optimization (MTO) algorithms against uniform equal weighting (EW), tuned static linear scalarization (LS), and isolated STL baselines. The candidates span dynamic loss weighting and gradient surgery. The evaluation covers Tagalog-English (Taglish) disaster tweets from four historical Philippine typhoons under a 4-fold leave-one-event-out (LOEO) protocol. Using PyTorch gradient diagnostic hooks and an intra-task half-batch baseline (Elich et al., 2024), the framework quantifies gradient interference and evaluates out-of-distribution generalization. The fine-tuned model is then deployed in a prototype web application for disaster triage.

1.1 Background of the Study

Disaster triage extracts three types of information from incoming crisis messages, namely location entities, humanitarian intent categories (Alam et al., 2021), and urgency levels (Wang et al., 2021). Deploying independent Transformer models for each task multiplies inference latency and parameter storage linearly as $\mathcal{O}(K)$ for K tasks. In contrast, hard parameter sharing passes text through a shared multilingual Transformer encoder to compute intermediate contextual representations, which feed lightweight task-specific linear heads (S. Chen et al., 2024; Ruder, 2017).

This unified architecture maintains near-constant inference latency ($\mathcal{O}(1)$ relative to K) on local workstation hardware. The shared encoder also introduces an inductive bias, because auxiliary tasks steer the model toward representations that explain more than one objective (Ruder, 2017).

Despite these architectural advantages, joint multi-task training introduces complex multi-objective optimization dynamics across shared parameters (Sener & Koltun, 2018). When task loss gradients point in opposing directions ($\langle \mathbf{g}_i, \mathbf{g}_j \rangle < 0$), updating shared parameters along task i increases the empirical loss on task j to a first-order approximation. This condition produces destructive interference and negative transfer across shared layers (B. Liu et al., 2021; Navon et al., 2022; Yu et al., 2020). Furthermore, when gradient scales diverge, tasks with larger gradient magnitudes dominate parameter updates and suppress secondary objectives (Z. Chen et al., 2018). Joint models of token-level entities and sequence-level intent have improved performance on both tasks (Weld et al., 2022). Whether such gains survive competing task gradients in a shared encoder remains an open empirical question.

Two primary MTO approaches address these optimization bottlenecks, namely dynamic loss weighting and gradient surgery. Dynamic loss-weighting algorithms adjust scalar task loss weights during training using distinct heuristics. Uncertainty weighting scales losses by learned homoscedastic uncertainty (Kendall et al., 2018). GradNorm balances gradient magnitudes against relative training paces (Z. Chen et al., 2018). Fast Adaptive Multitask Optimization (FAMO) equalizes task loss decrease rates (B. Liu et al., 2023). In contrast, gradient-surgery algorithms manipulate gradient vectors directly. Projecting Conflicting Gradients (PCGrad) projects conflicting gradients onto orthogonal normal planes (Yu et al., 2020). Conflict-Averse Gradient Descent (CAGrad) optimizes the worst-case objective improvement within a trust region (B. Liu et al., 2021). Impartial Multi-Task Learning (IMTL-G) enforces equal-angle descent across objectives (L. Liu et al., 2021). Nash-MTL formulates gradient aggregation as a multi-objective bargaining game (Navon et al., 2022).

Despite these theoretical motivations, empirical studies have shown conflicting evidence regarding whether dynamic MTO methods consistently outperform tuned LS or EW baselines. On machine translation and computer vision benchmarks, dynamic task weights often remained nearly constant during training, whereas per-task gradient computations reduced training throughput (Xin et al., 2022). Furthermore, under adaptive optimizers such as Adam, gradient magnitude disparities dominated multi-task updates, rendering vector surgery less effective and making EW competitive (Elich et al., 2024). Because modern Transformer fine-tuning relies on AdamW, this study investigates whether these findings hold on low-resource code-switched text with asymmetric token-level and sequence-level objectives. To separate genuine cross-task conflict from stochastic mini-batch noise, the experimental framework computes pairwise gradient angles against an intra-task half-batch baseline (Elich et al., 2024).

In the Philippines, disaster response represents a critical operational priority. The Philippine Area of Responsibility experiences an average of 20 tropical cyclones each year, about eight or nine of which cross the country (Philippine Atmospheric, Geophysical and Astronomical Services Administration, n.d.). During major typhoon events, citizens broadcast large volumes of distress messages across social media. Prominent events include Typhoon Haiyan (2013), Typhoon Vamco (2020), Typhoon Rai (2021), and Typhoon Paeng (2022). Filipino disaster response has likewise adopted unified hashtags such as #RescuePH and #ReliefPH (Ermino et al., 2022). Government agencies such as the National Disaster Risk Reduction and Management Council and the Office of Civil Defense monitor these streams. However, locating urgent calls by manually scanning typhoon-related posts is less efficient than automated triage (Ermino et al., 2022).

Taglish introduces severe linguistic complexity into automated triage. Taglish crisis messages combine Tagalog and English in single sentences, along with misspellings, abbreviations, and slang (Herrera et al., 2022; Montalan et al., 2025). Formations such as "ma-evacuate" and "nag-collapse" illustrate Tagalog affixes applied to English roots. Multilingual Transformer tokenizers segment these morphologically hybrid words into fragmented subword sequences. For Tagalog, the examined tokenizers required between 1.74 and 3.28 subword pieces per word (Petrov et al., 2023). This fragmentation inflates sequence length, dilutes semantic embeddings, and elevates gradient variance across shared layers. On other low-resource, code-switched text, joint training of sequence-labeling and classification tasks helped only in specific task configurations and did not benefit every task (Adouane & Bernardy, 2020).

Operational constraints in Philippine disaster management further dictate deployment requirements. Local government units and municipal command centers frequently operate on commodity workstations subject to power grid failures and telecommunication outages during typhoons. Relying on cloud-hosted large language model APIs introduces critical risks. These risks include transmission latency, service loss during internet blackouts, recurring API costs, and compliance obligations under the Philippine Data Privacy Act of 2012 (Republic of the Philippines, 2012). Deploying compact, fine-tuned multi-task models locally provides a resilient, self-contained triage capability.

Table 1

Structural Profile of Related Disaster Informatics and Philippine NLP Corpora

Dataset or Benchmark	Domain and Coverage	Modality and Corpus Volume	Language and Register	Annotation Level and Target Tasks	Cross-Event Evaluation Protocol
HumAID (Alam et al., 2021)	Global crisis informatics (19 disaster events)	Text only (77,196 tweets)	Monolingual English (social media)	Sequence-level (10 humanitarian intent categories)	Event-stratified and cross-event splits
Batayan (Montalan et al., 2025)	General Philippine NLP benchmark	Text only (3,800 test instances)	Tagalog and Taglish (diverse sources)	Task-isolated evaluations across eight NLP tasks	Task-specific benchmark splits
TweetTaglish (Herrera et al., 2022)	Code-switching analysis	Text only (21,150 tweets)	Informal Taglish (social media)	Document-level language mixing distribution	Random 80/20 train/test split
TLUNIFIED-NER (Miranda, 2023)	News and reference corpus	Text only (news articles)	Formal Tagalog (news articles)	Token-level sequence labeling (BIO entity spans)	Standard entity evaluation splits
Proposed Multi-Task Corpus	Philippine disaster triage (4 typhoon events)	Text only (6,000 to 10,000 tweets)	Informal Taglish (social media)	Concurrent token-level BIO spans, 4-class intent, and 3-tier urgency	4-fold LOEO

Note. NLP = natural language processing. BIO = beginning, inside, outside. LOEO = leave-one-event-out.

As summarized in Table 1, existing benchmarks exhibit clear architectural limitations. Global crisis corpora such as HumAID provide large-scale intent annotations but remain restricted to monolingual English and lack token-level entity spans. Philippine NLP benchmarks, such as Batayan, TweetTaglish, and TLUNIFIED-NER, evaluate single tasks in isolation or focus on formal news text. No existing public corpus provides concurrent annotations for token-level named entities, sequence-level intent, and sequence-level urgency on Taglish crisis messages under cross-event evaluation.

The absence of such a corpus directly impairs multi-task optimization research for crisis triage. In a shared Transformer encoder, token-level NER aggregates cross-entropy loss across all subword tokens in a sequence, whereas intent and urgency classification computes loss over a single pooled representation. Token-level cross-entropy therefore sums over many more positions than pooled classification, so its gradients can exceed those of the intent and urgency heads in magnitude. GradNorm targets exactly such magnitude disparities across tasks (Z. Chen et al., 2018). Under EW, this structural asymmetry may allow token-level updates to overwrite pooled sequence representations, inducing negative transfer.

To address these empirical and optimization challenges, this study curates the Multi-Task Corpus of Taglish Disaster Tweets across four Philippine typhoons under a 4-fold LOEO protocol. The study evaluates dynamic loss weighting and gradient surgery against baseline methods, providing empirical guidance and an operational triage artifact for crisis informatics.

1.2 Statement of the Problem

This study addresses the general problem of accurate disaster triage for Taglish crisis messages on local workstation hardware. A separate single-task model for each task scales latency and memory linearly with task count ($\mathcal{O}(K)$ for K tasks). A shared multilingual Transformer encoder reduces this cost to a single forward pass ($\mathcal{O}(1)$ relative to K). Joint training, however, couples heterogeneous task losses, and their gradients may differ in magnitude or point in opposing directions (Z. Chen et al., 2018; Yu et al., 2020). Such disparities and conflicts can induce negative transfer, leaving a joint model weaker than its single-task counterpart. Empirical evidence on these remedies remains conflicting. On translation and vision benchmarks, dynamic task weights stayed nearly constant and per-task gradient computations slowed training (Xin et al., 2022). Under adaptive optimizers such as Adam, magnitude disparities dominated updates and kept EW competitive, while stochastic mini-batch noise can be mistaken for genuine cross-task conflict (Elich et al., 2024). Finally, no public corpus pairs token-level and sequence-level triage annotations on Taglish crisis text under cross-event evaluation. The study therefore investigates whether principled multi-task optimization, grounded in a curated Taglish disaster corpus, yields a triage model that is accurate, statistically validated, and deployable on local hardware.

To align the investigation with the DSRM lifecycle, the study decomposes this general problem into four specific problems:

1. **Specific Problem 1 (Phase 1, Baseline Analysis).** What predictive F1 scores, execution latency, throughput, and memory footprint do isolated single-task learning models establish? How do these baselines compare with a shared multilingual encoder under EW across 4-fold LOEO splits in accuracy, gradient norm ratios ($R_{\text{max/min}}$), and pairwise gradient cosines?
2. **Specific Problem 2 (Phase 2, Design and Artifact Creation).** How should the hard-parameter-sharing pipeline, the candidate MTO algorithms, and the gradient diagnostic routines be designed to limit interference on Taglish text? How can the

resulting model be integrated into a prototype web application for disaster triage?

3. **Specific Problem 3 (Phase 3, Empirical Benchmarking).** How do the seven candidate MTO algorithms, spanning dynamic loss weighting and gradient surgery, compare with the STL, EW, and LS baselines across 4-fold LOEO splits? Which algorithms deliver the highest F1 scores, relative transfer (Δ_m), and computational efficiency, and the shortest training runtime?
4. **Specific Problem 4 (Phase 4, Comparative Validation).** Are the observed performance differences among the MTO algorithms statistically significant under two-tier nonparametric testing with paired bootstrap confidence intervals and Friedman–Nemenyi tests? How strongly does relative transfer (Δ_m) correlate with gradient conflict metrics measured against the intra-task half-batch baseline (Elich et al., 2024)?

1.3 Objectives of the Study

The general objective of this study is to construct the Multi-Task Corpus of Taglish Disaster Tweets and to benchmark candidate MTO algorithms on that corpus under 4-fold LOEO splits. The study further deploys the fine-tuned model in a prototype web application for disaster triage.

Aligned with these specific problems, the study pursues the following specific objectives:

1. **Specific Objective 1 (Phase 1, Baseline Analysis).** To establish the predictive F1 scores, execution latency, throughput, and memory footprint of isolated single-task learning models. To compare these baselines with a shared multilingual encoder

under EW across 4-fold LOEO splits in accuracy, gradient norm ratios ($R_{\text{max/min}}$), and pairwise gradient cosines.

2. **Specific Objective 2 (Phase 2, Design and Artifact Creation).** To design the hard-parameter-sharing pipeline, the candidate MTO algorithms, and the gradient diagnostic routines to limit interference on Taglish text. To integrate the resulting model into a prototype web application for disaster triage.
3. **Specific Objective 3 (Phase 3, Empirical Benchmarking).** To benchmark the seven candidate MTO algorithms, spanning dynamic loss weighting and gradient surgery, against the STL, EW, and LS baselines across 4-fold LOEO splits. To identify the algorithms delivering the highest F1 scores, relative transfer (Δ_m), and computational efficiency, and the shortest training runtime.
4. **Specific Objective 4 (Phase 4, Comparative Validation).** To test the observed performance differences among the MTO algorithms for statistical significance under two-tier nonparametric testing with paired bootstrap confidence intervals and Friedman–Nemenyi tests. To analyze the Spearman rank correlation (ρ) between relative transfer (Δ_m) and gradient conflict metrics measured against the intra-task half-batch baseline (Elich et al., 2024).

1.4 Theoretical Framework

This study rests on three theoretical foundations, namely DSRM, complexity analysis of shared representations, and multi-objective optimization with gradient dynamics.

DSRM treats research as the creation and evaluation of artifacts that solve identified organizational problems. Hevner et al. (2004) required that such research produce a viable artifact, address relevant problems, apply rigorous methods, and yield clear, verifiable contributions. Peffers et al. (2007) formalized this inquiry as an iterative lifecycle of six activities. These are problem identification and motivation; definition of the objectives for a solution; design and development; demonstration; evaluation; and communication. The study operationalizes this lifecycle as four sequential phases, namely Baseline Analysis (Phase 1), Design and Artifact Creation (Phase 2), Empirical Benchmarking (Phase 3), and Comparative Validation (Phase 4). Under this foundation, the intellectual contribution consists of the empirical optimization findings and the curated corpus, while the prototype serves as the demonstration artifact.

The second foundation combines asymptotic complexity analysis with the representation-sharing and inductive-bias principles described by Ruder (2017). For K tasks, single-task deployment executes K forward passes per input, for a total cost of $\mathcal{O}(K \cdot T_{\text{sh}})$, where T_{sh} is the backbone forward-pass time. Memory follows the same pattern, holding K separate encoders at $\mathcal{O}(K \cdot |\boldsymbol{\theta}_{\text{sh}}|)$ parameters, where $|\boldsymbol{\theta}_{\text{sh}}|$ is the encoder parameter count. Hard parameter sharing computes the shared representation once, reducing forward-pass time to

$$\mathcal{O}\left(T_{\text{sh}} + \sum_{k=1}^{K} T_{\text{head},k}\right)$$

where $T_{\text{head},k}$ denotes the head forward-pass time for task k. The linear heads are far cheaper than the backbone, so this total is effectively constant in K ($\mathcal{O}(1)$ relative to K). Parameter memory likewise falls to

$$\mathcal{O}\left(|\boldsymbol{\theta}_{\text{sh}}| + \sum_{k=1}^{K} |\boldsymbol{\theta}_{\text{head},k}|\right)$$

where $|\boldsymbol{\theta}_{\text{head},k}|$ denotes the parameter count of the head for task k. The heads are linear layers with negligible parameter volume, so memory is also effectively constant in K. Beyond cost, the shared encoder restricts the parameter search to representations that explain more than one task, the inductive bias that auxiliary tasks provide.

The third foundation casts joint training as multi-objective optimization over the shared parameters $\boldsymbol{\theta}_{\text{sh}}$ and the task heads $\boldsymbol{\theta}_{\text{head},1}, \ldots, \boldsymbol{\theta}_{\text{head},K}$ (Sener & Koltun, 2018):

$$\min_{\boldsymbol{\theta}_{\text{sh}}, \boldsymbol{\theta}_{\text{head},1}, \ldots, \boldsymbol{\theta}_{\text{head},K}} \left(\mathcal{L}_1(\boldsymbol{\theta}_{\text{sh}}, \boldsymbol{\theta}_{\text{head},1}), \ldots, \mathcal{L}_K(\boldsymbol{\theta}_{\text{sh}}, \boldsymbol{\theta}_{\text{head},K})\right)$$

This formulation seeks Pareto-optimal configurations where no task loss can decrease without increasing another. Let $\mathbf{g}_k \equiv \nabla_{\boldsymbol{\theta}_{\text{sh}}} \mathcal{L}_k$ denote the gradient of task k's loss with respect to the shared parameters. With learning rate η and local smoothness, a first-order expansion of task j's loss under a step along task i yields

$$\mathcal{L}_j(\boldsymbol{\theta}_{\text{sh}} - \eta \mathbf{g}_i, \boldsymbol{\theta}_{\text{head},j}) = \mathcal{L}_j(\boldsymbol{\theta}_{\text{sh}}, \boldsymbol{\theta}_{\text{head},j}) - \eta \langle \mathbf{g}_i, \mathbf{g}_j \rangle + \mathcal{O}\left(\eta^2 \|\mathbf{g}_i\|_2^2\right)$$

Yu et al. (2020) used this decomposition to analyze when updates help or harm other tasks. Two failure modes follow.

Disparity in Gradient Magnitudes. When tasks label tokens and classify sequences at once, gradient magnitudes can diverge, and the larger gradient dominates the update, leaving smaller-gradient tasks behind (Z. Chen et al., 2018).

Gradient Directional Conflict. When the inner product is negative ($\langle \mathbf{g}_i, \mathbf{g}_j \rangle < 0$), the first-order term becomes positive, so a step along task i raises the loss of task j, producing destructive interference (Yu et al., 2020).

Section 1.1 introduced the two MTO families that address these failure modes. Following the diagnostic principle of Elich et al. (2024), the framework compares pairwise gradient angles with an intra-task half-batch baseline to separate genuine conflict from mini-batch noise. Per-fold paired bootstrap confidence intervals form the first validation tier. For the second tier, the Friedman test with its Nemenyi post-hoc procedure covers multi-model comparisons (Demšar, 2006). Both tiers are nonparametric. To translate these principles into an operational workflow, the study adopts the Input-Process-Output (IPO) conceptual framework.

References

Adouane, W., & Bernardy, J.-P. (2020). When is multi-task learning beneficial for low-resource noisy code-switched user-generated Algerian texts? *Proceedings of the 4th Workshop on Computational Approaches to Code Switching*. https://aclanthology.org/2020.calcs-1.3

Alam, F., Qazi, U., Imran, M., & Ofli, F. (2021). HumAID: Human-annotated disaster incidents data from Twitter with deep learning benchmarks. *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM 2021)*. https://doi.org/10.1609/icwsm.v15i1.18116

Chen, S., Zhang, Y., & Yang, Q. (2024). Multi-task learning in natural language processing: An overview. *ACM Computing Surveys*. https://doi.org/10.1145/3663363

Chen, Z., Badrinarayanan, V., Lee, C.-Y., & Rabinovich, A. (2018). GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks. *Proceedings of the 35th International Conference on Machine Learning (ICML 2018)*. https://arxiv.org/abs/1711.02257

Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*. https://www.jmlr.org/papers/v7/demsar06a.html

Elich, C., Kirchdorfer, L., Köhler, J. M., & Schott, L. (2024). Examining common paradigms in multi-task learning. *46th DAGM German Conference on Pattern Recognition (GCPR 2024)*. https://doi.org/10.1007/978-3-031-85181-0/_9

Ermino, R. P., Abacan, R. C. A., Diamante, N. G. C., Faca, K. M. N., & Juntereal, T. E. D. (2022). Evaluating urgency of Typhoon-Related tweets through sentiment analysis using artificial neural networks. *International Journal of Computing Sciences Research*. https://doi.org/10.25147/ijcsr.2017.001.1.80

Herrera, M., Aich, A., & Parde, N. (2022). TweetTaglish: A dataset for investigating Tagalog-English code-switching. *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. https://aclanthology.org/2022.lrec-1.225

Hevner, A. R., Ram, S., March, S. T., & Park, J. (2004). Design science in information systems research. *MIS Quarterly*. https://doi.org/10.2307/25148625

Imran, M., Mitra, P., & Castillo, C. (2016). Twitter as a lifeline: Human-annotated Twitter corpora for NLP of crisis-related messages. *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC 2016)*.

https://doi.org/10.63317/43bfut6ipc3j

Kendall, A., Gal, Y., & Cipolla, R. (2018). Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2018)*. https://doi.org/10.1109/CVPR.2018.00781

Liu, B., Feng, Y., Stone, P., & Liu, Q. (2023). FAMO: Fast adaptive multitask optimization. *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. https://arxiv.org/abs/2306.03792

Liu, B., Liu, X., Jin, X., Stone, P., & Liu, Q. (2021). Conflict-averse gradient descent for multi-task learning. *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*. https://arxiv.org/abs/2110.14048

Liu, L., Li, Y., Kuang, Z., Xue, J.-H., Chen, Y., Yang, W., Liao, Q., & Zhang, W. (2021). TOWARDS IMPARTIAL MULTI-TASK LEARNING. *International Conference on Learning Representations*. https://openreview.net/forum?id=IMPnRXEWpvr

Miranda, L. J. V. (2023). Developing a named entity recognition dataset for Tagalog. *Proceedings of the First Workshop in South East Asian Language Processing*. https://doi.org/10.18653/v1/2023.sealp-1.2

Montalan, J. R., Layacan, J. P., Africa, D. D., Flores, R. I., Lopez II, M. T., Magsajo, T. D., Cayabyab, A., & Tjhi, W. C. (2025). Batayan: A Filipino NLP benchmark for evaluating large language models. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. https://doi.org/10.18653/v1/2025.acl-long.1509

Navon, A., Shamsian, A., Achituve, I., Maron, H., Kawaguchi, K., Chechik, G., & Fetaya,

E. (2022). Multi-task learning as a bargaining game. *International Conference on Machine Learning (ICML 2022)*. https://arxiv.org/abs/2202.01017

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems*. https://doi.org/10.2753/MIS0742-1222240302

Petrov, A., La Malfa, E., Torr, P. H. S., & Bibi, A. (2023). Language model tokenizers introduce unfairness between languages. *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. https://arxiv.org/abs/2305.15425

Philippine Atmospheric, Geophysical and Astronomical Services Administration. (n.d.). *Tropical cyclone information*. PAGASA. https://www.pagasa.dost.gov.ph/climate/tropical-cyclone-information

Republic of the Philippines. (2012). *Republic act no. 10173: Data privacy act of 2012*. Official Gazette. https://www.officialgazette.gov.ph/2012/08/15/republic-act-no-10173/

Ruder, S. (2017). *An overview of multi-task learning in deep neural networks*. arXiv. https://arxiv.org/abs/1706.05098

Sener, O., & Koltun, V. (2018). Multi-task learning as multi-objective optimization. *Advances in Neural Information Processing Systems 31 (NeurIPS 2018)*. https://arxiv.org/abs/1810.04650

Wang, C., Nulty, P., & Lillis, D. (2021). Transformer-based multi-task learning for disaster tweet categorisation. *Proceedings of the 18th International Conference on Information Systems for Crisis Response and Management (ISCRAM 2021)*. https://arxiv.org/abs/2110.08010

Weld, H., Huang, X., Long, S., Poon, J., & Han, S. C. (2022). A survey of joint intent detection and slot-filling models in natural language understanding. *ACM Computing Surveys*. https://doi.org/10.1145/3547138

Xin, D., Ghorbani, B., Garg, A., Firat, O., & Gilmer, J. (2022). Do current multi-task optimization methods in deep learning even help? *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*. https://arxiv.org/abs/2209.11379

Yu, T., Kumar, S., Gupta, A., Levine, S., Hausman, K., & Finn, C. (2020). Gradient surgery for multi-task learning. *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*. https://arxiv.org/abs/2001.06782